Supplementary Material for RelationalRAG-Tables: A Learned Join Verifier for QA over Tables and Knowledge Graphs

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1 Scope of this Supplement

This document collects the *supporting* stress tests behind the main paper's regime characterization. The main paper is self-contained, and nothing here is required to follow its argument. We gather these tables for two reasons: to give a table anchor to numbers the main paper reports inline, and to expose per-seed and per-regime detail that does not fit the page budget. Every number is a measured result from the experiment harness or the supplied post-adjudication correction snapshot; we report seeds explicitly and add no value we did not run. Section and table references of the form "main paper, Table N" point to the primary submission.

2 Baseline Definitions

For reproducibility we collect the exact definition of every baseline compared against the learned verifier. Each baseline scores the *same* retrieved candidate pairs as the verifier and differs only in the matching function; unless noted, a baseline accepts the highest-scoring candidate.

- **Exact** (exact code match): accept only if the candidate identifier is byte-identical to the table key. **Canonicalized exact** first lowercases and strips punctuation and whitespace; on our data it equals exact, confirming the drift is genuine rather than a formatting artifact.
- **Prefix / hierarchy rule**: accept the candidate whose code is the longest prefix of the key (or vice versa), the standard ICD category-to-subcode containment. Near-optimal on hierarchical drift, it collapses when no prefix relation holds (cross-vocabulary or corrupted codes).
- **Fuzzy (edit)**: rank by normalized edit similarity $1 - \mathrm{lev}(a, b)/\max(|a|, |b|)$; this is the deterministic fuzzy matcher used as an oracle in the synthetic study.
- **Jaro–Winkler** (Winkler, 1990): rank by Jaro–Winkler string similarity with default parameters.
- **Character TF–IDF** (Sung et al., 2020): cosine similarity over character n-gram TF–IDF vectors of the two identifiers.
- **Embedding** ("Embed"): cosine similarity over description embeddings. On the synthetic core these are the reference implementation's deterministic feature-hash embeddings (no pretrained model); on the real-data layer they are a pinned pretrained sentence-transformer whose exact checkpoint and version are recorded in the reproduction guide.
- **Fine-tuned cross-encoder**: a MiniLM cross-encoder (Li et al., 2020) fine-tuned, Ditto-style, to read code and description jointly, trained on the same positive (gold) and hard-negative (near-duplicate identifier) pairs as the verifier (Section 7).
- **Fellegi–Sunter** (Fellegi and Sunter, 1969): the classical probabilistic record-linkage log-likelihood-ratio matcher; the full estimator is given in Section 16 and the main paper.
- **No-verifier (NV)**: the planner commits the top-ranked retrieved candidate without the verifier gate, isolating what gating prevents.
- **Oracle**: uses gold join labels; an upper bound, not a deployable method.
- **Learned verifier** (ours): the MLP over the explicit identifier-pair feature vector of the main paper, optionally a K-member deep ensemble with an uncertainty head.

3 Main-Text Tables Replaced by Plots

The main paper promotes five dense result tables into compact figures. Tables 1–4 and Table 5 preserve the exact numeric values behind those plots.

Table 1: End-to-end exact match by verifier and task family on the drifted synthetic corpus (1,392 questions, 60% join drift). "Abst." is the number of abstentions. Oracle is an upper bound with gold join labels.

Verifier	Single	Join	Multi	Back	Abst$_t$	EM	Abst.
Exact-match	1.00	0.43	0.43	0.43	**1.00**	0.569	636
No verifier	1.00	**1.00**	**1.00**	0.40	0.00	0.860	0
Learned	1.00	0.93	0.93	**0.58**	0.08	0.856	50
Deep ensemble	1.00	0.94	0.94	0.57	0.08	**0.861**	47
Oracle (UB)	1.00	1.00	1.00	1.00	1.00	1.000	36

Table 2: Heterogeneous-drift join (one ICD-10 KG, each record drifting by hierarchy, OCR-corruption, *or* cross-vocabulary), recall@1, mean over five seeds with sd as gray subscript.

Method	recall@1
Exact code	$0.011_{\pm 0.009}$
Fuzzy (edit)	$0.251_{\pm 0.023}$
Prefix / hierarchy rule	$0.258_{\pm 0.030}$
Embedding	$0.689_{\pm 0.035}$
Learned verifier	$0.752_{\pm 0.035}$

4 ICD-10 + Wikidata: OCR-Corruption Sweep

The main paper states that under attempted data-entry corruption the prefix-hierarchy rule collapses while embedding on the surviving description overtakes it. Table 6 sweeps the probability of calling the corruption routine at a single seed. The audit found that some calls were no-ops, so this axis is the *configured attempt rate*, not the realized fraction of invalid codes. The observed prefix recall falls $0.729 \rightarrow 0.397$ while embedding is flat; the verifier never significantly beats the prefix rule ($p > 0.45$). We retain this as an implementation stress test and do not use it to estimate subtype-specific performance.

5 Heterogeneous Drift: Per-Seed Detail

The main paper reports aggregate performance on a column generated with hierarchy, OCR-attempt, and cross-vocabulary mechanisms. Table 7 gives the five-seed breakdown behind that mean. The verifier beats the prefix rule decisively in all five seeds ($\Delta \approx +0.49$, $p < 10^{-3}$ throughout) and beats embedding in three of five. Relabeling no-op OCR attempts changes no code, gold node, feature, prediction, or aggregate score; we omit the unsupported pre-adjudication per-subtype breakdown.

Table 3: Alias-corrected BIRD value linking: recall@1 over five held-out splits (mean ± sd; 387 test instances per split).

Scorer	recall@1 (all)	recall@1 (drift)
Exact match	0.808 ± 0.012	0.000 ± 0.000
Fuzzy match	0.666 ± 0.015	0.167 ± 0.038
Embedding cosine	0.860 ± 0.011	0.806 ± 0.031
Learned verifier	$\mathbf{0.939 \pm 0.010}$	$\mathbf{0.905 \pm 0.051}$

Table 4: End-to-end Execution Accuracy on BIRD dev (local 7B-Coder).

Pipeline	EX	EX (drift)
Ablation (n=800, 72 drift)		
baseline (full schema)	0.458	0.292
+ column values	0.476	0.333
+ value hints (all)	0.446	0.347
+ value hints (selective)	0.456	0.319
+ schema pruning	0.479	0.319
+ pruning + values	**0.483**	**0.361**
Full dev (n=1534, 131 drift)		
baseline	0.454	0.328
+ pruning + values	**0.477**	**0.420**

6 Robustness to Description Noise

We retain the robustness probe that does not depend on the audited subtype labels. Under 50% word-dropout of descriptions (Table 8), the verifier's drift recall@1 falls $0.746 \rightarrow 0.712$ while embedding falls $0.458 \rightarrow 0.424$, and the prefix rule is unaffected because it never reads the description. The pre-adjudication hierarchy-to-OCR transfer result is omitted because no-op edits contaminated its subtype partition.

7 A Stronger Matcher: Fine-Tuned Cross-Encoder

A natural worry is that the feature-MLP verifier is too weak to settle the "learned vs. rule" question. We replace it with a fine-tuned MiniLM cross-encoder that reads code and description jointly (Table 9). On the original BIRD extraction it reaches 0.95 overall and on drift, but on HybridQA it loses to lexical matching ($0.205 < 0.235$), and on the ICD hierarchy task (main paper) it still ties the prefix rule ($p \approx 0.89$). Because this expensive training run predates the BIRD alias correction, we retain it only as an exploratory capacity check; the corrected feature-verifier result in the main paper carries the BIRD claim. The cross-regime negative result still shows that capacity alone is not sufficient.

Table 5: Cross-vocabulary join (ICD-9 table → ICD-10 KG, 129 held-out), recall@1.

Method	recall@1
Exact code	0.008
Fuzzy code	0.008
Prefix rule	0.008
Embedding	**0.907**
Learned verifier	0.899

Table 6: ICD-10 join, *drift-subset* recall@1 as a function of the configured code-corruption attempt rate (seed 0, with 59 to 69 drifted test records). Some attempts are no-ops; p is a paired bootstrap of verifier vs. prefix.

Corrupt frac.	Exact	Fuzzy	Prefix	Embed	Verifier
0.0 (clean)	0.000	0.644	**0.729**	0.458	0.746
0.5	0.015	0.493	**0.594**	0.493	0.580
1.0 (full)	0.032	0.365	0.397	**0.460**	0.413
Verifier vs. prefix: n.s. throughout (p=0.59, 0.86, 0.45)					

Table 7: Heterogeneous-drift join, aggregate per-seed recall@1 (one ICD-10 KG, with each record assigned a hierarchy, OCR-attempt, or cross-vocabulary mechanism). Δ and p are verifier vs. embedding (paired bootstrap, 10^4 resamples).

Seed	Exact	Fuzzy	Prefix	Embed	Verifier	Δ_{emb}	p
0	0.022	0.256	0.261	0.678	0.700	+0.022	0.257
1	0.006	0.228	0.256	0.733	0.794	+0.061	0.016
2	0.022	0.283	0.311	0.672	0.772	+0.100	<0.001
3	0.000	0.222	0.222	0.639	0.772	+0.133	<0.001
4	0.006	0.267	0.239	0.722	0.722	+0.000	0.532
mean	0.011	0.251	0.258	0.689	**0.752**	+0.063	3/5 sig.

Table 8: Description-noise sensitivity: drift-subset recall@1 under 50% word-dropout of record and node descriptions (seed 0).

Method	clean desc.	noisy (50% drop)
Prefix / hierarchy rule	0.729	0.729
Embedding	0.458	0.424
Learned verifier	**0.746**	**0.712**

8 HybridQA: Full Retrieval and Reranking

HybridQA is the main paper's negative result: with gold cell-to-passage links and no identifier drift, locating the answer-bearing unit is lexically dominated. Table 10 gives the retrieval curve. Lexical overlap dominates dense `bge-small` retrieval at every cutoff (recall@20 0.826 vs. 0.647), and a learned evidence verifier, although it improves binary discrimination (area under the ROC curve, AUROC, 0.610 vs. 0.505 and 0.493 for the dense and lexical signals alone), *loses* at top-1 reranking (0.091 vs. lexical's 0.249). String matching is the right tool here, and we report it.

9 BIRD Value Linking: Discrimination and Selective Prediction

Beyond the recall@1 results in the main paper, the alias-corrected verifier remains near-perfect as a pair classifier. Across five seeds it reaches recall@1 0.939 ± 0.010 overall and 0.905 ± 0.051 on drift. On the held-out seed-4 split (387 test, 44 drifted), recall@1 is 0.951 overall and 0.955 on drift, AUROC is 0.995, and selective prediction raises accuracy on the answered fraction from 0.994 at full coverage to 1.000 at 50% (Table 11).

10 End-to-End Execution Accuracy: Scale and Full Dev

The main paper's end-to-end gain holds at full-dev scale and at larger model size (Table 12). At full dev (n=1534) the schema-pruning-plus-values pipeline lifts EX 0.454→0.477 (+2.3), with the drift subset gaining +9.2 (0.328→0.420). A 14B coder improves identically in relative terms (+2.3 EX), confirming the gain is not an artifact of the 7B model. Absolute EX stays below cloud-scale systems, so the contribution is the relative effect and the schema-linking diagnosis.

11 Calibration, System-Level Selective Prediction, and Retrieval Noise

These diagnostics answer questions the controlled study did not originally report: calibration error, selective prediction at the level of QA answers rather than pairwise joins, and the value of gating once retrieval is no longer idealized.

End-to-end selective prediction. Beyond pairwise joins, we rank each answered QA question by the uncertainty of its committed join and defer the most-uncertain tail (Table 13). System-level exact match rises from 0.889 at full coverage to 1.000 at 50% coverage. The curve is noisier than the pairwise one, because entropy is a weak error predictor on the aleatoric joins and because deterministic single-table lookups occupy the high-confidence region, but the endpoints give a clean system-level accuracy-coverage operating curve.

Retrieval noise. The synthetic study places the gold node in the candidate pool, which inflates the no-verifier

Table 9: Fine-tuned cross-encoder (MiniLM) recall@1.

Task (held-out)	Baseline	Cross-encoder
BIRD value linking, all (387)	n/a	**0.954**
BIRD value linking, drift (38)	n/a	**0.947**
HybridQA top-1 (395)	0.235 (lexical)	0.205

Table 10: HybridQA retrieval recall@k ($\sim$397 dev questions). Top-1 reranking and verifier AUROC are reported in the text.

Retrieval	@1	@5	@10	@20
Dense (`bge-small`)	0.129	0.390	0.511	0.647
Lexical	**0.244**	**0.544**	**0.670**	**0.826**
Hybrid	0.234	0.474	0.647	0.778

baseline. We stress this on bad-join-backtrack questions, where retrieval surfaces a plausible wrong neighbor, by dropping the gold candidate with probability p (Table 14). The no-verifier planner commits the wrong neighbor and returns a silently wrong answer for 0.60 of questions at every p, and it never abstains. The verifier holds the wrong-answer rate lower and converts the growing pool of unrecoverable joins into explicit abstentions, whose rate climbs from 0.02 at p=0 to 0.29 at p=1. The value of gating grows precisely as retrieval degrades, which aggregate exact match on idealized retrieval hides. The p=0 row reconciles exactly with the backtrack column of the main paper (no-verifier EM 0.40, ensemble 0.57).

Faithfulness. On the same QA run, answers are *faithful* (grounded in retrieved evidence or a deterministic traversal) for 0.94 of non-abstained questions. The deterministic planner cannot fabricate, and the residual gap is retrieval coverage on single-table lookups rather than hallucination.

Cost-aware utility. Accuracy-coverage curves understate the deployment value of abstention, so we score a correct answer +1, a wrong answer $-\lambda$, and an abstention 0, and find the utility-maximizing coverage (Table 15). When wrong answers are cheap (λ=1) the optimum answers everything. When a wrong answer costs five times a correct one (λ=5), the high-stakes regime the verifier targets, the optimum is to abstain on the most-uncertain 40% of answers, which raises utility from 0.33 at full coverage to 0.60 at coverage 0.6. Abstention pays off precisely when errors are costly.

Table 11: Alias-corrected BIRD value linking, held-out (seed 4). Pair-classification AUROC by scorer, and ensemble selective-prediction accuracy vs. coverage.

AUROC	verifier	exact	fuzzy	embed-cos	
	0.995	0.938	0.909	0.829	
Coverage	1.00	0.90	0.80	0.70	0.50
Accuracy	0.994	0.999	1.000	1.000	**1.000**

Table 12: End-to-end Execution Accuracy (BIRD dev, local coder).

Setting (n)	EX base	EX +p+v	Δ	Δ drift
7B, full dev (1534)	0.454	**0.477**	+2.3	+9.2
14B (400)	0.490	**0.513**	+2.3	+6.7

12 Threshold Sensitivity, Compute-Matched Uncertainty, KG Incompleteness, and Efficiency

These diagnostics cover threshold stability, whether the deep ensemble earns its compute, robustness to graph incompleteness, a deployment-oriented latency breakdown, and the cost of gating.

Threshold sensitivity. Table 16 sweeps the acceptance threshold τ_v. Precision rises and recall falls monotonically, and F1 is flat (0.75 to 0.77) across $\tau_v \in [0.4, 0.6]$, so the fixed τ_v=0.5 sits in a stable band near the F1 optimum rather than on a cliff.

Does the ensemble earn its compute? We compare the five-member ensemble against two single-model signals at one-fifth the compute: the entropy of one member and a score-margin heuristic, which rank identically (Table 17). For *selective prediction* the single model matches the ensemble (both reach 0.86 at 50% coverage, and the single model is in fact slightly higher at full coverage, mirroring its higher raw accuracy in the main paper). The ensemble's value is therefore *calibration* (ECE 0.027 vs. the single model's 0.036, in the main-text calibration table), not selective-prediction ranking, which is consistent with the saturating K-sweep in the main paper. A practitioner who needs only ranking can use a single model.

KG incompleteness. Removing a fraction of graph edges (Table 18) lowers exact match as expected, but the lost answers become *abstentions*, not wrong answers: at 10% and 20% edge removal the wrong-answer rate stays at 0.02 and 0.04 while abstention rises to 0.13 and 0.41. The verifier-gated planner degrades gracefully under graph staleness, refusing rather than fabricating.

Table 13: End-to-end (system-level) selective prediction: exact match on the answered fraction of QA questions after deferring the most-uncertain committed joins (1,345 answered).

Coverage	1.00	0.90	0.80	0.70	0.50
System EM	0.889	0.890	0.876	0.913	**1.000**

Table 14: Retrieval-noise robustness on bad-join-backtrack questions (264 questions), dropping the gold candidate with probability p. NV = no-verifier, Ens = ensemble.

	No-verifier			Ensemble		
p	EM	wrong	abst	EM	wrong	abst
0.00	0.40	0.60	0.00	**0.57**	0.41	0.02
0.25	0.40	0.60	0.00	0.51	0.42	0.06
0.50	0.40	0.60	0.00	0.43	0.44	0.13
0.75	0.40	0.60	0.00	0.34	0.45	0.20
1.00	0.40	0.60	0.00	0.25	0.46	**0.29**

Efficiency. Table 19 breaks per-question latency into stages over 1,392 questions on a 6,509-entry index. Retrieval dominates (P50/P95 10.7/11.6 ms), and execution, which includes the verifier and ensemble scoring, is sub-millisecond (P50/P95 0.6/1.2 ms). Verification is not the bottleneck at this scale, and the production path for larger corpora is approximate search.

Where gating hurts. The gate is not free. On clean join questions the no-verifier planner is perfect (EM 1.00) because the gold candidate is present, whereas the verifier rejects 16 of 264 correct joins as false negatives near the decision boundary, lowering EM to 0.94 (a 6% cost). This is the price of safety on easy positives, and HybridQA (main paper) is the macro case where gating loses outright. Raising τ_v trades this cost against bad-join rejection.

13 Verifier Feature Ablation

A natural question is whether the learned verifier merely recovers one obvious rule, such as numeric-suffix agreement. We retrain it on single-feature subsets. On the synthetic graded-corruption split (Table 20) no subset matches the full model: digit agreement reaches AUROC 0.70, edit-ratio 0.76, prefix overlap 0.67, and embedding only 0.52, while the full fusion reaches 0.82. The verifier learns a combination, not a single cue, and on this digit-preserving generator even digit agreement alone is far from sufficient. On the real ICD-10 hierarchy task (Table 21) the picture differs and matches the main paper: the prefix and code-structure features alone already reach the verifier's recall@1 (0.826/0.729 all/drift), because hierarchy drift *is* a prefix relation, and the embedding adds only a small drift gain (full 0.746). Where a structural shortcut exists a single feature suffices, and where it does not the fusion is necessary.

Table 15: Cost-aware utility on the synthetic QA set. A correct answer scores $+1$, a wrong answer $-\lambda$, and an abstention 0. Abstention becomes utility-optimal as wrong answers grow costly.

Wrong-answer penalty λ	Best coverage	Utility (best / full)
1	1.0	0.78 / 0.78
2	1.0	0.67 / 0.67
5	0.6	0.60 / 0.33

Table 16: Verifier discrimination on the held-out split as the acceptance threshold τ_v varies.

τ_v	0.3	0.4	0.5	0.6	0.7
Precision	0.657	0.696	0.767	0.808	0.883
Recall	0.923	0.869	0.738	0.702	0.500
F1	0.768	**0.773**	0.752	0.751	0.639

14 Verifier Failure Analysis

Table 22 characterizes where the verifier errs. On the alias-corrected BIRD seed-0 split, the false-positive rate is 0.46% and the false-negative rate 5.43%. On the ICD-10 join the dominant error is different and more instructive. Of the 41 recall@1 misses on 236 test records (17%), 18 are *sibling-code confusions* where the chosen wrong node shares the gold's three-character category, exactly the hard case (`R508` vs. `R509`) that a prefix rule also struggles with. The verifier moreover *commits* most of these (38 of 41 score above τ_v) rather than abstaining, and it rejects the gold pair in 18% of cases. This is the candid weakness behind the tie against the prefix rule: near-duplicate hierarchical codes are where both methods fail, and the verifier is overconfident there.

15 Scalability of the Multi-Index Retriever

The full scalability table is reported in the main paper. As an additional note on that study, the pure-Python linear-scan retriever reaches a P50 latency of 206 ms at $\sim$ 104k index entries, up from 11 ms at 6.5k entries—a 19× rise for a 16× corpus—while the verifier's per-pair cost stays constant. Retrieval, not verification, is therefore the scaling bottleneck, and the crossover near 10^5 entries is where the production path (approximate nearest-neighbor search and index sharding) replaces the exact scan.

Table 17: Compute-matched selective prediction: accuracy on the answered fraction, ensemble (5×) vs. a single-model signal (1×).

Coverage	1.00	0.90	0.80	0.70	0.50
Ensemble (5×)	0.725	0.757	0.783	0.803	0.856
Single model (1×)	0.743	0.755	0.779	0.793	0.861

Table 18: KG incompleteness on join and multi-hop questions (792 questions).

Edges removed	0%	10%	20%	30%
Exact match	0.939	0.845	0.552	0.518
Wrong	0.010	0.023	0.035	0.046
Abstain	0.051	0.133	0.413	0.437

16 Record-Linkage Baseline, Composite Keys, Structural Context, and Risk-Controlled Selective Prediction

We add four analyses. The record-linkage baseline, composite-key, and risk-controlled selective-prediction studies run on the deterministic synthetic core. The structural-context study runs on the real ICD-10 layer, where the code hierarchy supplies genuine 1-hop context.

Classical record-linkage baseline. We benchmark a Fellegi–Sunter probabilistic matcher (Fellegi and Sunter, 1969), the canonical non-neural record-linkage model, on the same features and pairs as the learned verifier. We binarize each join feature into a field-agreement indicator, estimate the agreement probability under a match (m) and a non-match (u) from the labelled training pairs with Laplace smoothing, and score a pair by the summed log-likelihood ratio

$$s_{\mathrm{FS}} = \sum_j \mathbb{1}[a_j] \log \frac{m_j}{u_j} + \mathbb{1}[\neg a_j] \log \frac{1-m_j}{1-u_j},$$

where a_j denotes agreement on field j. Fellegi–Sunter is a strong, training-light baseline (AUROC 0.753), and the learned MLP verifier on the identical features improves on it by +0.067 AUROC; the headline comparison is reported in the main paper. The gain is real but modest, which is the honest reading. A learned matcher buys interaction structure over the continuous features that the agreement-binarized likelihood ratio discards, and the strongest agreement fields it recovers (containment, full-digit, and digit-suffix match) are exactly the numeric-key signals the generator preserves.

Table 19: Stage-wise per-question latency (CPU, 1,392 questions, 6,509 index entries).

Stage	P50 (ms)	P95 (ms)
Retrieval	10.7	11.6
Execution (incl. verifier)	0.6	1.2
Total	11.3	15.2

Table 20: Synthetic verifier trained on feature subsets, held-out discrimination. No single feature matches the full model.

Feature subset	Accuracy	AUROC
Digit agreement	0.678	0.699
Edit ratio	0.683	0.757
Prefix overlap	0.637	0.669
Embedding	0.566	0.517
Full	**0.745**	**0.820**

Composite (multi-attribute) keys: AND/OR composition and conflicts. The verifier scores one pair, but a composite join over a two-attribute key is scored by combining the per-attribute match probabilities. A probabilistic AND (their product, accepting above τ_v^2) suits keys where both components must agree, and a probabilistic OR $(1-(1-s_1)(1-s_2)$, accepting above τ_v) suits alternate keys where either component identifies the record. We evaluate positives (both attributes drift yet agree), conflicts (one attribute agrees, one does not), and easy negatives (neither agrees). The product ranks composite matches well (AUROC 0.934), far above single-attribute scoring (AUROC 0.737), which cannot tell a positive from a conflict because it never inspects the disagreeing attribute (Table 23). The two operators encode opposite key semantics on the conflict cases, and this is the central point: AND rejects a conflict (accept rate 0.018), as a composite key should, while OR treats one matching component as sufficient (accept rate 0.60 on conflicts, recall 0.91 on positives), as a many-to-one or alternate key should. Both recover drifted positives that exact composite matching almost entirely misses (0.3% recall). The composition operator is therefore a modeling choice that encodes the key's semantics, not a fixed recipe, and the per-attribute uncertainties compose into a join-level abstention signal either way.

Risk-controlled selective prediction. A natural question is whether conformal or risk-controlling selective classification (Geifman and El-Yaniv, 2017; Angelopoulos and Bates, 2023) should replace deep-ensemble abstention. We find the two are complementary rather than competing. On a held-out calibration split we select, for each target risk α, the largest accepted set whose calibration error is at most α, then report the achieved coverage and error on a disjoint test split (Table 24). The realized test error tracks the target, holding at $\alpha \in \{0.02, 0.05\}$

Table 21: ICD-10 verifier trained on feature subsets, recall@1. The structural (prefix/code) features carry the hierarchy signal, and the embedding adds a small drift gain.

Feature subset	recall@1 (all)	recall@1 (drift)
Prefix overlap	0.826	0.729
Edit ratio	0.805	0.644
Embedding	0.754	0.458
Code structure	0.826	0.729
Full	**0.826**	**0.746**

Table 22: Verifier failure analysis. BIRD errors are conservative, while ICD errors are dominated by sibling-code confusion that the verifier commits rather than abstains on.

ICD-10 join (236 test records, recall@1)	
recall@1 misses	41 (17%)
sibling-code confusion	18
committed a wrong join	38
safely abstained	3
gold pair rejected (FN)	42 (18%)
BIRD value linking (held-out pairs, τ_v=0.5)	
false negatives (gold rejected)	21 (5.43%)
false positives (wrong accepted)	75 (0.46%)

with a small finite-sample overshoot at $\alpha = 0.10$. Conformalizing the *ensemble* score rather than a single member lowers the realized risk at the strictest target (0.007 versus 0.013 at $\alpha = 0.01$) and buys more coverage at the loosest (0.424 versus 0.399 at $\alpha = 0.10$). Conformal selection supplies the calibrated operating point that the ensemble's heuristic entropy threshold lacks, and the ensemble supplies the better-ranked confidence score that conformal selection thresholds. The low coverage at strict targets is itself informative. Guaranteeing a sub-5% join error under heavy drift means accepting only the most confident quarter of pairs and abstaining on the rest, which is precisely the safety-coverage trade-off the verifier exists to expose.

Lightweight structural context. We also ask whether adding lightweight structural context beyond the pairwise features improves discrimination or calibration without sacrificing speed. We test this on the real ICD-10 layer, where the code hierarchy supplies genuine structure: the shared three-character category (the 1-hop parent in the ICD tree), the shared chapter letter, code-length specificity agreement, and a sibling indicator. We compare the base six-feature verifier against the base plus these four structural features on a controlled split, where the training pairs and ensemble seeds are identical and only the feature map differs. Structural context improves both discrimination and calibration at no measurable cost: AUROC rises from 0.834 to 0.855

Table 23: Composite two-attribute key joins under drift, with AND (product) and OR composition. AND rejects conflicts (one component disagrees), as a composite key should, while OR accepts a single matching component, as an alternate or many-to-one key should. Both recover drifted positives that exact composite matching misses.

Method	AUROC	Recall$^+$	Conflict accept
Exact composite	n/a	0.003	0.000
Single-attribute verifier	0.737	n/a	n/a
Per-attribute OR	0.886	0.910	0.600
Per-attribute AND (product)	**0.934**	0.428	0.018

Table 24: Split-conformal selective prediction. Coverage and achieved test risk at each target risk α, conformalizing a single member versus the full ensemble. Realized risk tracks α, and the ensemble score gives tighter risk at strict targets and more coverage at loose ones.

	Single member		Ensemble	
Target α	Cov.	Risk	Cov.	Risk
0.01	0.212	0.013	0.197	0.007
0.02	0.212	0.013	0.197	0.007
0.05	0.244	0.044	0.236	0.034
0.10	0.399	0.110	0.424	0.113

and expected calibration error falls from 0.064 to 0.039, while per-pair feature extraction stays at about 6 μs (the structural features are constant-time string operations), Table 25. Top-1 recall barely moves (+0.4 points) and sibling-code confusions drop by only one. The honest reading is that 1-hop hierarchy context sharpens the score and its calibration but does not resolve sibling confusion, because siblings share the very category the context encodes, so disambiguating them needs finer structure than a single hop. This answers the question in the affirmative for calibration and ranking, and it scopes the open part precisely.

Table 25: Lightweight structural context (ICD-10 hierarchy) added to the pairwise verifier, controlled split. Discrimination (AUROC) and calibration (ECE) improve at no latency cost, while top-1 recall and sibling confusion barely move because siblings share the encoded category.

Verifier	AUROC	ECE	R@1 (all/drift)	Sib. conf.	μs/pair
Base (6 features)	0.834	0.064	0.822/0.746	19	6.2
+ structural (10)	**0.855**	**0.039**	0.826/0.746	18	6.0

17 Normalization Baselines, Recalibration, Regime Detection, and Robustness

We add further analyses.

Practical normalization and fuzzy baselines. The consolidated ICD baselines in the main text add three matchers common in data engineering. Domain-aware code canonicalization (stripping dots and spacing, then exact match) scores exactly as plain exact (0.644 all, 0.000 drift), which shows the ICD drift is genuine hierarchy drift, not a formatting artifact that a normalization pipeline would remove. Jaro–Winkler (Winkler, 1990) on canonicalized codes ties the prefix rule (0.826 all, 0.729 drift), confirming that a strong off-the-shelf fuzzy matcher matches the hand-crafted rule on homogeneous hierarchy drift. Character TF-IDF cosine trails embedding (0.729 all, 0.407 drift). The learned verifier's drift-subset edge (0.746) is therefore over fairly-tuned string baselines.

Single-model recalibration. Because ensemble ranking does not beat a single model, we ask whether temperature scaling or isotonic regression on one model recovers the ensemble's calibration at one-fifth the compute. It does not (Table 26). Fitting temperature on a held-out calibration half leaves test ECE at 0.039, since the single model already sits near temperature 1, and isotonic regression overfits the calibration half to 0.043. Both remain above the ensemble's 0.024 on the same split, so the ensemble's calibration advantage is not recovered by a cheap post-hoc fix here.

Table 26: Single-model recalibration versus the deep ensemble: test-half ECE on the synthetic split. Neither temperature scaling nor isotonic regression closes the gap to the ensemble.

Method	ECE
Single, raw	0.039
Single, temperature scaled (T=1.1)	0.039
Single, isotonic	0.043
Deep ensemble	**0.024**

Threshold stability on real data. The acceptance threshold is stable on real ICD pairs, not only in the synthetic study (Table 27). As τ_v rises from 0.3 to 0.7, precision rises and recall falls monotonically, and accept-decision F1 stays within $[0.79, 0.80]$ for $\tau_v \in [0.4, 0.6]$, so the fixed τ_v=0.5 sits in a stable band rather than on a cliff, matching the synthetic threshold sweep.

Table 27: Planner-threshold sensitivity on real ICD pairs: precision, recall, and F1 of the accept decision as τ_v varies. F1 is stable near the operating point.

τ_v	0.3	0.4	0.5	0.6	0.7
Precision	0.732	0.745	0.759	0.829	0.835
Recall	0.881	0.856	0.839	0.758	0.708
F1	**0.800**	0.797	0.797	0.792	0.766

Regime detector. A per-instance regime detector is feasible, which is why the cascade's failure is the verifier's cross-regime miscalibration rather than the routing decision. A logistic classifier over six cheap surface features (token count, digit and alphabetic fractions, mean token length, code punctuation, and a single-token flag) separates identifier queries from natural-language spans, using ICD codes as identifiers and their descriptions as spans, at 0.997 five-fold cross-validated accuracy against a 0.5 majority baseline. Routing on inputs is therefore easy; the open problem is a verifier whose score is calibrated across regimes.

Split protocol and leakage. The ICD train and test sets split on records. We quantify residual overlap on the held-out test records: only 1.3% share their exact billable code with a training record and 19.5% share their gold Wikidata node, while 41.1% share the coarse three-character category. The verifier cannot exploit category-level signal because it reads only relational features of a (record, node) pair and never node identity, so the modest category overlap is not a leakage path.

Real-data verifier training. We elaborate how the real-data verifier pairs are built, since the protocol governs both leakage and generalization. For each training record we form one positive pair, (record, gold node), and eight negatives by *hard-negative mining*: the four KG nodes whose label-plus-code embedding is closest to the record (the distractors a retriever would actually surface) and four uniformly random nodes. Each ensemble member trains on a bootstrap resample of these pairs with a distinct hidden width, and labels are taken as given rather than smoothed, since the ICD pairing is exact at construction (a record's gold node is its true category). The deliberate label ambiguity is confined to the *synthetic* graded-corruption split, where the same-entity probability decreases with edit distance through a $\sim$50/50 boundary at two edits, which is what gives the verifier a non-degenerate operating point and the ensemble meaningful variance.

Retrieval noise on real data. The retrieval-noise probe of the controlled study repeats on the real ICD corpus (Table 28). We drop the gold KG node from the candidate pool with probability p and compare the no-verifier planner, which commits the top-scored node, against the verifier, which commits only when the top score clears τ_v. As p rises the no-verifier wrong-answer rate climbs to 1.00, because every dropped-gold record yields a silent wrong join, whereas the verifier holds its wrong rate near 0.25 and converts the rest into abstentions, reaching 0.75 at p=1. The value of gating again grows as retrieval degrades, now on real identifiers.

Enterprise-scale retrieval. We demonstrate the scaling cost model empirically rather than asserting it (Ta-

Table 28: Retrieval noise on the real ICD corpus: dropping the gold node with probability p. The verifier converts unrecoverable joins into abstentions while the no-verifier planner returns silent wrong answers.

p	0.00	0.25	0.50	0.75	1.00
No-verifier wrong	0.18	0.39	0.59	0.79	1.00
Verifier wrong	0.18	0.18	0.21	0.24	0.25
Verifier abstain	0.00	0.21	0.39	0.56	**0.75**

ble 29). Over N unit 384-vectors we time a FAISS exact (flat) index and an approximate inverted-file (IVF) index. Exact search grows linearly, from 0.04 ms at 10^4 to 28 ms at 10^6 candidates, while approximate search stays under half a millisecond throughout, roughly a 60× gap at 10^6 (latencies are mean per-query and hardware-dependent, but the linear-versus-flat trend is the robust point). Because the verifier scores only the few candidates the index returns, its per-join cost is constant in N, so at enterprise scale retrieval, not verification, is the bottleneck and approximate search keeps it cheap.

Table 29: Mean per-query latency of FAISS exact (flat) versus approximate (IVF) search over N unit 384-vectors. Exact search scales linearly with N; approximate search stays roughly flat.

N	10^4	10^5	10^6
Exact (flat) ms	0.036	0.344	28.041
Approximate (IVF) ms	0.016	0.012	**0.480**

18 Additional Studies

The main paper reports six further studies whose headline rows appear there. Original outputs are in the artifact (`artifacts/multiseed`, `theory_check.json`, `e1`, `icd_kg/results_e5_cascade.json`, `e6`, `e7`), each regenerated by a named driver; the supplied annotation package adds the post-adjudication alias-correction driver and snapshot used for the corrected BIRD rows.

Multi-seed synthetic study (E2). `relational_rag_tables.multiseed` regenerates the corpus and retrains the verifiers for 20 seeds and writes per-seed JSON plus an aggregate with mean, sd, and t-based 95% CI for every scalar, 1%-step risk–coverage curves with 500-resample bootstrap bands, AURC and excess AURC, and validation-half threshold selection. The per-K selective-prediction rows, the retrieval-noise probe, the feature ablation, and KG incompleteness (Tables 17, 14, 20, 18) now live only here.

Theory check (E3). `relational_rag_tables.theory_check` measures α, β, π_+ on held-out pairs and r, r_0, δ on gold join pairs for every seed and compares the theorems' predictions with the planner's measured utility and join-task EM.

Table 30: Consolidated baselines on the ICD-10-CM → Wikidata join, recall@1 (all / drift). Canonicalization equals exact, confirming the drift is genuine hierarchy drift rather than formatting.

Method	All ↑	Drift ↑
Exact code match	0.644	0.000
Canonicalized exact	0.644	0.000
Fuzzy (edit) (Winkler, 1990)	0.805	0.644
Jaro–Winkler (Winkler, 1990)	0.826	0.729
Character TF-IDF (Sung et al., 2020)	0.729	0.407
Prefix / hierarchy rule	0.826	0.729
Embedding (Efthymiou et al., 2017)	0.754	0.458
Learned verifier (Mudgal et al., 2018; Li et al., 2020)	**0.826**	**0.746**

Further checks moved from the main paper. Per question, retrieval dominates cost (P50/P95 10.7/11.6 ms) and verification is sub-millisecond (0.6/1.2 ms); latency grows linearly with corpus size (11 → 206 ms at 104k entries) while per-pair cost stays constant, so approximate nearest-neighbour search takes over near 10^5 entries (Table 29). LLM matchers are excluded from the core comparison because the verifier runs sub-millisecond per pair against 100–500 ms per LLM call, the pipeline stays reproducible without API dependence, and the cross-encoder (Table 9) already shows that capacity is not the operative variable. Under 50% word-dropout of descriptions the verifier's drift recall@1 falls only 0.746 → 0.712 (Table 8); on the real ICD corpus, dropping the gold node with probability p drives the no-verifier wrong rate to 1.00 at p=1 while the verifier holds near 0.25 by abstaining (Table 28); a Fellegi–Sunter matcher on identical features reaches AUROC 0.753 against the verifier's 0.820.

Tables moved from the main paper. Table 30 gives the consolidated ICD-10 baselines. The schema-linker comparison (BIRD, n=800, with column values): full schema EX 0.476 (exec-err 0.131), embedding-only pruning 0.473 (0.141), learned pruning 0.483 (0.131), oracle gold tables 0.525 (0.114). Representative verifier decisions: `ACC-1001`→`ACX1001` scored 0.91 (correct join); `J45.9`→`J45.901` 0.87 (hierarchy match); `E11.65` vs. `E11.69` 0.81 (sibling confusion, committed); "high school"→`HS` 0.12 (correct reject).

Verifier-isolated BIRD ablation (E1). `relational_rag_real.run_e1` ran 16 configurations on a CUDA server with vLLM and the AWQ 4-bit coder: baseline, pruning + values without hints, five linkers, five drift-only-hint variants, and three stricter thresholds. Drift-only results: exact yields no hints; fuzzy hurts (−4.4 EX, precision 0.01); embedding −1.0; verifier −0.3

Table 31: Heterogeneous ICD drift, recall@1, mean over five seeds (sd as gray subscript): hand-tuned and linear fusions against the learned verifier. Rule-only rows are in Table 2.

Fusion	recall@1
Tuned cascade prefix→fuzzy→embedding	$0.776_{\pm 0.021}$
Logistic fusion (same features)	$0.771_{\pm 0.022}$
Learned verifier (MLP ensemble)	$0.757_{\pm 0.043}$
Learned verifier + pool-relative features	$0.771_{\pm 0.018}$

(+1.5 drift at τ=0.8); oracle drifted literals +0.5 EX, +6.1 drift. These are the submitted extractor's as-run results. The alias correction changes cross-fit training pools; the LLM generations were not repeated, although EX itself remains scored against the official gold SQL.

Regime-aware joint matcher (E4). `relational_rag_real.run_e4_joint` trains one 5-member MLP ensemble on pooled BIRD value-linking and HybridQA evidence pairs with common base, pool-relative, and regime features. On the original extraction it improves cross-regime calibration and closes two thirds of the HybridQA gap to lexical matching without beating it (Table 34). Because the run predates the BIRD alias correction, it is an exploratory routing diagnostic rather than corrected BIRD evidence.

Tuned rule cascades (E5). `relational_rag_real.run_e5_cascade` tunes cascade order and thresholds on the training split (prefix, then fuzzy at ≥ 0.9, then embedding in all five seeds). The aggregate comparison in Table 31 is unchanged by relabeling no-op OCR cases because subtype is not an input to any scorer. We omit the original per-subtype breakdown, which mixed valid hierarchy codes into the OCR stratum.

Label protocol and noise (E6). The supplied, self-contained `annotation/` package contains the two 100-item samples, completed independent A/B worksheets, offline evidence, provenance, integrity hashes, and standard-library agreement/adjudication scripts. The audit has 100 paired items per dataset: BIRD $\kappa = 0.81$ for protocol-gold correctness (four disagreements, all resolved by the gold SQL) and 1.00 for drift, with adjudicated error rates 0.12/0.00; ICD $\kappa = 1.00$ for drift type and undefined for gold correctness, with adjudicated rates 0.00/0.05. The package also includes the alias-corrected BIRD rerun and its machine-readable output; Table 33 uses the corrected BIRD extraction. The injected-noise sweep is reported as sensitivity evidence, not as a replacement for correction.

End-to-end table+KG QA (E7). `relational_rag_real.run_e7_icd_qa` builds the records table and the live-fetched Wikidata KG, writes the fixture in the synthetic corpus format (`fixtures/icd_qa/seed_{0..4}`), and runs the planner with each gate. Per seed: 436 records (21 ghosts), 159 test records, 518 questions (159 lookup, 167 join, 70 backtrack, 101 two-hop, 21 abstain), 2,126 triples. Per-task EM for the verifier gate: join 0.75, backtrack 0.72, two-hop 0.78, abstain 1.00; for the ungated planner: 0.90, 0.69, 0.87, 0.00.

19 Reproduction

Every table is tied to a machine-readable output and command. The artifact's reproduction guide maps the synthetic and real-data tables to their drivers; the annotation package's `recompute_bird_protocol_impact.py` applies the adjudicated alias correction and produces the corrected BIRD component and noise results without modifying the artifact checkout. The remaining real-data and synthetic tables regenerate as documented. Pretrained embeddings and the local coder make real-data runs seeded but not bit-identical across hardware, and reported values are single-seed point estimates except where a mean±sd over seeds is given. E1, E4, and the cross-encoder are explicitly marked as pre-correction exploratory or as-run analyses; they are not used for the corrected BIRD headline.

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Table 32: Measured inputs of Theorem 1 and Corollary 1 of the main paper (held-out pairs, τ_v=0.5), the implied critical penalty $\lambda^\star$, and how often the sign of the theorem's prediction matches the planner's measured utility at $\lambda \in \{1, 2, 5\}$.

Dataset	π_+	α (FPR)	β (FNR)	$\lambda^\star$	sign agrees
Synthetic (20 seeds)	$0.564_{\pm 0.011}$	$0.327_{\pm 0.045}$	$0.206_{\pm 0.038}$	$0.39_{\pm 0.05}$	20/20
BIRD value linking	0.023	0.0046	0.054	0.0013	—

Table 33: Label-noise sensitivity: a fraction ρ of training gold assignments is corrupted, the verifier retrained, and the clean test split scored. Gate rates at τ_v=0.5.

ρ	BIRD r@1 all / drift	BIRD FPR / FNR	ICD r@1 (3 seeds)	ICD FPR / FNR
0.00	0.928 / 0.947	0.46% / 5.4%	$0.752_{\pm 0.042}$	0.09% / 16.7%
0.05	0.920 / 0.895	0.47% / 4.7%	$0.739_{\pm 0.045}$	0.09% / 19.3%
0.10	0.910 / 0.868	0.42% / 5.7%	$0.726_{\pm 0.037}$	0.09% / 22.2%
0.20	0.910 / 0.895	0.26% / 25.6%	$0.733_{\pm 0.035}$	0.05% / 34.1%

Table 34: E4: one matcher for both regimes. Recall@1 (BIRD all / drift, HybridQA) and pair-level ECE (BIRD / HybridQA), three seeds.

Scorer	BIRD all	BIRD drift	HybridQA	ECE B / H
Lexical (edit ratio / coverage)	$0.710_{\pm 0.023}$	$0.381_{\pm 0.095}$	$0.281_{\pm 0.030}$	—
Dense cosine	$0.832_{\pm 0.025}$	$0.750_{\pm 0.058}$	$0.174_{\pm 0.023}$	—
Containment	$0.925_{\pm 0.009}$	$0.716_{\pm 0.092}$	$0.169_{\pm 0.012}$	—
BIRD-only verifier (24 feats)	$0.924_{\pm 0.006}$	$0.863_{\pm 0.048}$	$0.146_{\pm 0.043}$	0.001 / 0.117
HybridQA-only verifier (24 feats)	$0.343_{\pm 0.090}$	$0.419_{\pm 0.128}$	$0.229_{\pm 0.029}$	0.108 / 0.039
Joint, base + regime feats	$0.934_{\pm 0.006}$	$0.871_{\pm 0.055}$	$0.222_{\pm 0.030}$	0.002 / 0.026
Joint, + pool-relative feats	$0.924_{\pm 0.007}$	$0.863_{\pm 0.048}$	$0.237_{\pm 0.030}$	0.001 / 0.034

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